

## Problem Set 4

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1. a)

$$\text{logit}(p) = \alpha + \beta_1 X_1$$

where  $p$  is the probability of favorable response and  $\alpha$  is the log odds of favorable outcome when dose is 1 mg and  $\beta_1$  is the increment in log odds for 1 unit increase in  $\log_{10}(\text{dose})$  and  $X_1$  is  $\log_{10}(\text{dose})$ .

The assumptions are that we have a dichotomous response variable. The response is favorable or unfavorable pain relief. It is assumed that these data arise from a stratified simple random sample so that the favorable v. unfavorable pain relief is binomially distributed for each dose level.

b)

```
data hw4_1;
  input treatment $ dose response $ count @@;
  ldose=log10(dose);
  datalines;
  A 1 F 43 A 1 U 57
  A 10 F 55 A 10 U 45
  A 100 F 68 A 100 U 32
  B 2 F 34 B 2 U 66
  B 20 F 52 B 20 U 48
  B 200 F 75 B 200 U 25
  ;

proc print;
run;

*Checking data entry;

/* proc freq data = hw4_1; */
/*   tables treatment*dose*response / norow nocol nopercnt; */
/*   weight count; */
/* run; */

title "Problem 1";
proc logistic data=hw4_1;
  where treatment='A';
  weight count;
  model response(event='F')=ldose / scale=none aggregate;
```

```
run;
```

#### Deviance and Pearson Goodness-of-Fit Statistics

| Criterion | Value  | DF | Value/DF | Pr > ChiSq |
|-----------|--------|----|----------|------------|
| Deviance  | 0.0201 | 1  | 0.0201   | 0.8874     |
| Pearson   | 0.0201 | 1  | 0.0201   | 0.8874     |

Number of unique profiles: 3

If the model fits,  $Q_P$  is distributed as  $\chi^2_{(1)}$ . The null hypothesis is that the model fits the data.  $Q_P = 0.0201$  ( $p = 0.8874$ ). As this p-value is  $> 0.05$ , we would fail to reject the null hypothesis and conclude that the model fit is adequate.

c)

```
data hw4_1probit;
  input dose favb total @@;
  ldose=log10(dose);
  datalines;
1      43    100
10     55    100
100    68    100
;
proc print;
run;

*ods graphics on;
title2 "1c";
proc probit data=hw4_1probit log10 plots=none;
  model favb/total=dose / dist=logistic lackfit inversecl (prob=0.25 0.50 0.75);
run;
title2;
```

#### The Probit Procedure

##### Probit Analysis on Log10(dose)

| Probability | Log10(dose) | 95% Fiducial Limits |         |
|-------------|-------------|---------------------|---------|
| 0.25        | -1.5571     | -4.8023             | -0.5842 |
| 0.50        | 0.5675      | -0.2402             | 1.0183  |
| 0.75        | 2.6921      | 1.9720              | 4.9708  |

Problem 1

1c

#### The Probit Procedure

### Probit Analysis on dose

| Probability | dose      | 95% Fiducial Limits |          |
|-------------|-----------|---------------------|----------|
| 0.25        | 0.02773   | 0.0000158           | 0.26047  |
| 0.50        | 3.69395   | 0.57523             | 10.43028 |
| 0.75        | 492.10604 | 93.75945            | 93491    |

The ED25 is 0.02773 with confidence limits (0.000,0.260). The ED50 is 3.694 with confidence limits (0.575, 10.430). The ED75 is 492.106 with confidence limits (93.759, 93491).

d)

```

title2 "1d";
proc probit data=hw4_1probit log10 plot=ipppplot;
    model favb/total=dose / dist=normal lackfit inversec1 (prob=0.25 0.50 0.75);
run;

```

The Probit Procedure

### Probit Analysis on Log10(dose)

| Probability | Log10(dose) | 95% Fiducial Limits |         |
|-------------|-------------|---------------------|---------|
| 0.25        | -1.5273     | -4.6871             | -0.5649 |
| 0.50        | 0.5686      | -0.2388             | 1.0169  |
| 0.75        | 2.6646      | 1.9610              | 4.8472  |

Problem 1

1d

The Probit Procedure

### Probit Analysis on dose

| Probability | dose      | 95% Fiducial Limits |          |
|-------------|-----------|---------------------|----------|
| 0.25        | 0.02969   | 0.0000206           | 0.27233  |
| 0.50        | 3.70365   | 0.57708             | 10.39676 |
| 0.75        | 461.94059 | 91.41672            | 70346    |

The ED25 is 0.02969 with confidence limits (0.0000,0.2723). The ED50 is 3.7034 with confidence limits (0.5571, 10.39676). The ED75 is 461.9406 with confidence limits (91.4167,70346). The doses and confidence limits are almost identical to the ones calculated in 1c). However, the ED75 for logistic distribution was about 30 mg higher than the ED75 for normal distribution and the intervals for ED75 were not as close as the intervals for other dose levels.

2. a) The dilution assumption for doses  $z_a$  of Drug A and  $z_b$  of Drug B can be stated as

$$z_b = \rho z_a$$

Doses with comparable response for the two drugs are related by the constant  $\rho$ , the relative potency so that  $\rho$  mg of Drug B produce the same behavior of 1 mg of Drug A. It also implies that:

$$x_b = \log_{10}(\rho) + x_a$$

where  $x_a$  and  $x_b$  are the  $\log_{10}(\text{doses})$  of Drug A and Drug B respectively. We create separate models for Drug A and Drug B and we are modeling the probability of favorable response. Thus the logistic model structure for Drug B is:

$$p_b(x_{bi}) = \frac{1}{1 + \exp(-\alpha_b - \beta x_{bi})}$$

where  $x_{bi}$  denotes  $\log_{10}$  dose levels of Drug B. The logits are formed as follows:

$$\text{logit} \left\{ \frac{p_a(x_{ai})}{1 - p_a(x_{ai})} \right\} = \alpha_a + \beta x_{ai}$$

$$\text{logit} \left\{ \frac{p_b(x_{bi})}{1 - p_b(x_{bi})} \right\} = \alpha_b + \beta x_{bi}$$

b)

```
title 'Problem 2';

data hw4_2;
  set hw4_1;
  int_A=(treatment='A');
  int_B=(treatment='B');
run;

title2 "2b";

proc logistic data=hw4_2;
  weight count;
  model response=int_A int_B ldose*int_A ldose*int_B ldose*int_A*ldose*int_A
    ldose*int_B*ldose*int_B / noint scale=none aggregate include=4
    selection=forward details;
  contrast 'equal slopes' int_A*ldose 1 int_B*ldose -1;
run;
```

Deviance and Pearson Goodness-of-Fit Statistics

| Criterion | Value  | DF | Value/DF | Pr > ChiSq |
|-----------|--------|----|----------|------------|
| Deviance  | 0.3135 | 2  | 0.1568   | 0.8549     |
| Pearson   | 0.3137 | 2  | 0.1568   | 0.8549     |

Residual Chi-Square Test

|            |    |            |
|------------|----|------------|
| Chi-Square | DF | Pr > ChiSq |
| 0.3137     | 2  | 0.8549     |

#### Contrast Test Results

| Contrast     | DF | Wald<br>Chi-Square | Pr > ChiSq |
|--------------|----|--------------------|------------|
| equal slopes | 1  | 2.8211             | 0.0930     |

$Q_L = 0.3135$  and  $Q_P = 0.3137$ . Both p-values are 0.85, which supports adequate model fit. The residual score test for the quadratic terms was performed and had a value of 0.3137, which is also supportive of good model fit. The dilution assumption is done with a Wald test and tests the hypothesis  $H_0 : \beta_a = \beta_b$ . The Wald statistic was 2.28211 ( $p = 0.0930$ ). As this p-value is  $> 0.05$ , we would fail to reject the null hypothesis and conclude that a parallel lines model appears to fit these data.

c)

```
*Fitting parallel lines model;
title2 "2c";
proc logistic data=hw4_2 outest=estimate(drop=intercept _link_ _lnlike_) covout;
    freq count;
    model response=int_A int_B ldose / noint scale=none aggregate covb;
run;

title;
%Fieller(data=estimate, out=RP, num=1 -1 0, den=0 0 1,
    label=B:A log10(Rel. Potency), intercept=no);
```

| Ratio                   | Estimate | LCL       | UCL     |
|-------------------------|----------|-----------|---------|
| B:A log10(Rel. Potency) | 0.40582  | -0.092032 | 0.92159 |

Output from the Fieller macro is shown above. We exponentiate these values to get the actual values. Thus, a dose of drug B must be  $10^{0.4058} = 2.55$  times higher than a dose of Drug A to have the same effect with 95% confidence limits (0.8090, 8.3481).

3. a)

$$\begin{aligned} \log(\mu) = & \alpha + \beta_1(\text{age}='25-29') + \beta_2(\text{age}='30-34') \\ & + \beta_3(\text{age}='35-39') + \beta_4(\text{age}='40+') \\ & + \beta_5(\text{birth order}='2') + \beta_6(\text{birth order}='3') \\ & + \log\left(\frac{N}{100,000}\right) \end{aligned}$$

This is the regression equation for modeling the rate of disordered births and  $N$  represents the offset so the total number of births in each population group.  $\frac{\mu}{N} \times 100,000$  represents the rate of disorder per 100,000 live births.  $\beta_1$ - $\beta_6$  represent the parameters to be estimated.  $\exp(\alpha)$  = incidence rate of disorders per 100,000 live births when maternal age is 20-24 years and

birth order is 1

$\exp(\beta_1)$  = multiplicative effect on incidence rate per 100,000 live births due to maternal age = 25-29 years

$\exp(\beta_2)$  = multiplicative effect on incidence rate per 100,000 live births due to maternal age = 30-34 years

$\exp(\beta_3)$  = multiplicative effect on incidence rate per 100,000 live births due to maternal age = 35-39 years

$\exp(\beta_4)$  = multiplicative effect on incidence rate per 100,000 live births due to maternal age = 40+ years

$\exp(\beta_5)$  = multiplicative effect on incidence rate per 100,000 live births due to birth order = 2

$\exp(\beta_6)$  = multiplicative effect on incidence rate per 100,000 live births due to birth order = 3  
b)

```
title "Problem 3";
data hw4_3;
    input age $ birth_order $ cases total;
    total_n = total/100000;
    ltotal=log(total_n);
    datalines;
20-24 1 120 329472
20-24 2 151 326736
20-24 3 64 175688
25-29 1 50 114990
25-29 2 120 208690
25-29 3 107 207062
30-34 1 40 39455
30-34 2 85 83225
30-34 3 106 117334
35-39 1 38 14207
35-39 2 89 28490
35-39 3 102 45018
40+ 1 20 3053
40+ 2 44 5389
40+ 3 86 8655
;
run;

ods output diffs=birth_order_diffs;
proc genmod data=hw4_3 plots=none;
    class age(ref='20-24') birth_order(ref='1');
    model cases=age birth_order / dist=poisson link=log offset=ltotal;
    *estimate '2 vs 1 birth order' birth_order 1 0 / exp;
    *estimate '3 vs 1 birth order' birth_order 0 1 / exp;
    lsmeans birth_order / pdiff exp cl;
    output out=out(where=(age='40+')) p=pcount xbeta=xb stdxbeta=std;
run;
```

b)

# Analysis Of Maximum Likelihood Parameter Estimates

| Parameter   |       | DF | Estimate | Standard Error | Wald 95% Confidence Limits |        | Wald Chi-Square | Pr |
|-------------|-------|----|----------|----------------|----------------------------|--------|-----------------|----|
| Intercept   |       | 1  | 3.6037   | 0.0711         | 3.4643                     | 3.7431 | 2566.65         |    |
| age         | 25-29 | 1  | 0.2504   | 0.0823         | 0.0892                     | 0.4117 | 9.26            |    |
| age         | 30-34 | 1  | 0.8667   | 0.0875         | 0.6951                     | 1.0383 | 98.01           |    |
| age         | 35-39 | 1  | 1.8677   | 0.0880         | 1.6952                     | 2.0402 | 450.48          |    |
| age         | 40+   | 1  | 3.0820   | 0.1001         | 2.8859                     | 3.2782 | 948.61          |    |
| age         | 20-24 | 0  | 0.0000   | 0.0000         | 0.0000                     | 0.0000 | .               |    |
| birth_order | 2     | 1  | 0.1956   | 0.0767         | 0.0453                     | 0.3459 | 6.50            |    |
| birth_order | 3     | 1  | 0.0527   | 0.0796         | -0.1033                    | 0.2086 | 0.44            |    |
| birth_order | 1     | 0  | 0.0000   | 0.0000         | 0.0000                     | 0.0000 | .               |    |
| Scale       |       | 0  | 1.0000   | 0.0000         | 1.0000                     | 1.0000 |                 |    |

NOTE: The scale parameter was held fixed.

| Obs | Effect      | birth_order | _birth_order | Exponentiated | Exponentiated Lower | Exponentiated Upper |
|-----|-------------|-------------|--------------|---------------|---------------------|---------------------|
| 1   | birth_order | 2           | 3            | 1.1536        | 1.0141              | 1.3124              |
| 2   | birth_order | 2           | 1            | 1.2160        | 1.0463              | 1.4133              |
| 3   | birth_order | 3           | 1            | 1.0541        | 0.9019              | 1.2319              |

We can exponentiate the parameter estimates to get the incidence density ratios.  $e^{0.1956} = 1.216$  is the ratio of incidence of disorder for birth order 2 vs. birth order 1 with confidence limits (1.0463,1.4133).

$e^{0.0527} = 1.0541$  is the ratio of incidence of disorder for birth order 3 vs. birth order 1 with confidence limits (0.9019,1.2319).

c)

```
data predrates;
  set out;
  obsrate=cases/total;

  /* observed rate */
  lograte=xb-ltotal;
  prate=exp(lograte)

  /* predicted rate */
  lcl=exp(lograte-probit(.975)*std);
  ucl=exp(lograte+probit(.975)*std);
  keep total cases age birth_order prate lcl ucl;
run;

title2 "3c";
proc print data=predrates;
```

```

    id total cases;
run;
title;

```

| total | cases | age | birth_<br>order | prate   | lcl     | ucl     |
|-------|-------|-----|-----------------|---------|---------|---------|
| 3053  | 20    | 40+ | 1               | 800.890 | 656.858 | 976.51  |
| 5389  | 44    | 40+ | 2               | 973.898 | 815.890 | 1162.51 |
| 8655  | 86    | 40+ | 3               | 844.199 | 711.308 | 1001.92 |

SAS output is shown above for predicted rates of disorder per 100,000 live births in the 40+ age group. The predicted rate for birth order=1 is 800.890 disorders per 100,000 live births. The predicted rate for birth order=2 is 973.898 disorders per 100,000 live births. The predicted rate for birth order=3 is 844.199 disorders per 100,000 live births.

4.  
a)

| Interval            | No Caries | Caries | Withdrawals | At Risk |
|---------------------|-----------|--------|-------------|---------|
| <b>Existing</b>     |           |        |             |         |
| 0-6 months          | 125       | 3      | 2           | 130     |
| 6-12 months         | 118       | 5      | 2           | 125     |
| 12-18 months        | 97        | 16     | 5           | 118     |
| 18-24 months        | 50        | 35     | 12          | 97      |
| <b>Experimental</b> |           |        |             |         |
| 0-6 months          | 128       | 1      | 1           | 130     |
| 6-12 months         | 123       | 3      | 2           | 128     |
| 12-18 months        | 98        | 10     | 15          | 123     |
| 18-24 months        | 60        | 20     | 18          | 98      |

Table 1: Life Table Format of Caries Data

Define  $n_{ijk}$  to be the number of patients in the  $i$ th group with  $j$ th status for  $k$ th time interval, where

$i = 1, 2$  for existing and experimental treatment

$j = 0$  corresponds to no caries during the time interval

$j = 1, 2$  corresponds to those with caries and those withdrawn during  $k$ th interval, respectively

$k = 1, \dots, t$  time intervals

The  $n_{i0k}$  are determined from:

$$n_{i0k} = \sum_{j=1}^2 \sum_{g=k+1}^t n_{ijg} + n_{i0t}$$

b)

| Lower<br>STRATUM | Upper<br>treatment | Time | Time | Survival | StdErr |
|------------------|--------------------|------|------|----------|--------|
|------------------|--------------------|------|------|----------|--------|



|   |       |   |   |        |         |
|---|-------|---|---|--------|---------|
| 1 | exist | 0 | 1 | 1.0000 | 0       |
| 1 | exist | 1 | 2 | 0.9767 | 0.0133  |
| 1 | exist | 2 | 3 | 0.9374 | 0.0214  |
| 1 | exist | 3 | 4 | 0.8075 | 0.0353  |
| 1 | exist | 4 | . | 0.4969 | 0.0466  |
| 2 | exper | 0 | 1 | 1.0000 | 0       |
| 2 | exper | 1 | 2 | 0.9923 | 0.00769 |
| 2 | exper | 2 | 3 | 0.9688 | 0.0153  |
| 2 | exper | 3 | 4 | 0.8850 | 0.0290  |
| 2 | exper | 4 | . | 0.6861 | 0.0451  |

c)

$$G_{11} = \frac{125 + 0.5(2)}{125 + 3 + 0.5(2)} = 0.9767$$

$$G_{12} = 0.9767 \times \frac{118 + 0.5(2)}{118 + 5 + 0.5(2)} = 0.9374$$

$$G_{13} = 0.9374 \times \frac{97 + 0.5(5)}{97 + 16 + 0.5(5)} = 0.8075$$

$$G_{14} = 0.8075 \times \frac{50 + 0.5(12)}{50 + 35 + 0.5(12)} = 0.4969$$

d)

Cochran-Mantel-Haenszel Statistics (Based on Table Scores)

| Statistic | Alternative Hypothesis | DF | Value  | Prob   |
|-----------|------------------------|----|--------|--------|
| 1         | Nonzero Correlation    | 1  | 7.2951 | 0.0069 |
| 2         | Row Mean Scores Differ | 1  | 7.2951 | 0.0069 |
| 3         | General Association    | 1  | 7.2951 | 0.0069 |

The null hypothesis is that the survival rates are the same for each treatment group.  $Q_{MC} = 7.2951$  ( $p = 0.0069$ ). As  $p$  is  $< 0.05$ , we reject the null and conclude that there is a significant treatment effect on survival.

5. a) The assumptions for piecewise exponential model are as follows:

The withdrawals are uniformly distributed during the time intervals in which they occur.

They are unrelated to treatment failures and do not have the event at the time of withdrawal.

The within-interval probabilities of the treatment failures are small and have independent exponential distributions.

Piecewise exponential model assumes independent exponential distributions with hazard parameters  $\lambda_{ik}$  for respective time periods.

$\alpha$  = log incidence density for existing treatment, 0-6 months

$\eta_1$  = incremental effect for 6-12 months

$\eta_2$  = incremental effect for 12-18 months

$\eta_3$  = incremental effect for 18-24 months  
 $\beta$  = incremental effect for existing treatment  
The model is specified as follows:

$$\lambda_{ik} = \exp(\alpha + \eta_k + \mathbf{x}_i' \beta_k)$$

$\{\alpha + \eta_k\}$  is the constant value of the log hazard function within the  $k$ th interval when  $x_i = 0$  for  $i = 1$  corresponds to experimental treatment and  $i = 2$  corresponds to existing treatment.  $\beta$  relates the hazard function for the  $i$ th population to the explanatory variables  $x_i$ .  
b)

\*\*\*\*\*

#### Problem 5

\*\*\*\*\*;

```
data hw4_5;
input treatment $ time $ failure months;
nmonths=log(months);
datalines;
exist _0-6 3 765
exist 6-12 5 729
exist 12-18 16 645
exist 18-24 35 441
exper _0_6 1 774
exper 6-12 3 753
exper 12-18 10 663
exper 18-24 20 474
;
proc print;
run;

proc genmod data = hw4_5;
    class treatment(ref='exist') time;
    model failure = time treatment / d=poisson link=log offset=nmonths;
run;
```

#### Analysis Of Maximum Likelihood Parameter Estimates

| Parameter |       | DF | Estimate | Standard Error | Wald 95% Confidence Limits |         | Wald Chi-Square | Pr >   |
|-----------|-------|----|----------|----------------|----------------------------|---------|-----------------|--------|
| Intercept |       | 1  | -6.0671  | 1.0238         | -8.0737                    | -4.0605 | 35.12           | <.0001 |
| time      | 12-18 | 1  | 2.4030   | 1.0286         | 0.3871                     | 4.4189  | 5.46            | 0.0201 |
| time      | 18-24 | 1  | 3.5160   | 1.0183         | 1.5200                     | 5.5119  | 11.92           | <.0001 |
| time      | 6-12  | 1  | 1.1002   | 1.0698         | -0.9966                    | 3.1969  | 1.06            | 0.3074 |
| time      | _0-6  | 1  | 0.5259   | 1.1754         | -1.7778                    | 2.8295  | 0.20            | 0.6529 |
| time      | _0_6  | 0  | 0.0000   | 0.0000         | 0.0000                     | 0.0000  | .               | >.0001 |
| treatment | exper | 1  | -0.5844  | 0.2195         | -1.0146                    | -0.1543 | 7.09            | 0.0077 |
| treatment | exist | 0  | 0.0000   | 0.0000         | 0.0000                     | 0.0000  | .               | >.0001 |
| Scale     |       | 0  | 1.0000   | 0.0000         | 1.0000                     | 1.0000  |                 |        |

The hazard ratio is calculated by exponentiating so we have  $e^{\hat{\beta}} = e^{-0.5844} = 0.5574$  with confidence limits (0.3625,0.8570). The hazard of caries in experimental treatment group is 0.5574 times the hazard of caries in the existing treatment group.

c) The failure rate per month for existing treatment within 6-12 months time interval can be calculated as follows:

$$\exp(\hat{\alpha} + \hat{\eta}_1) = \exp(-6.60671 + 1.1002) = 0.00696$$